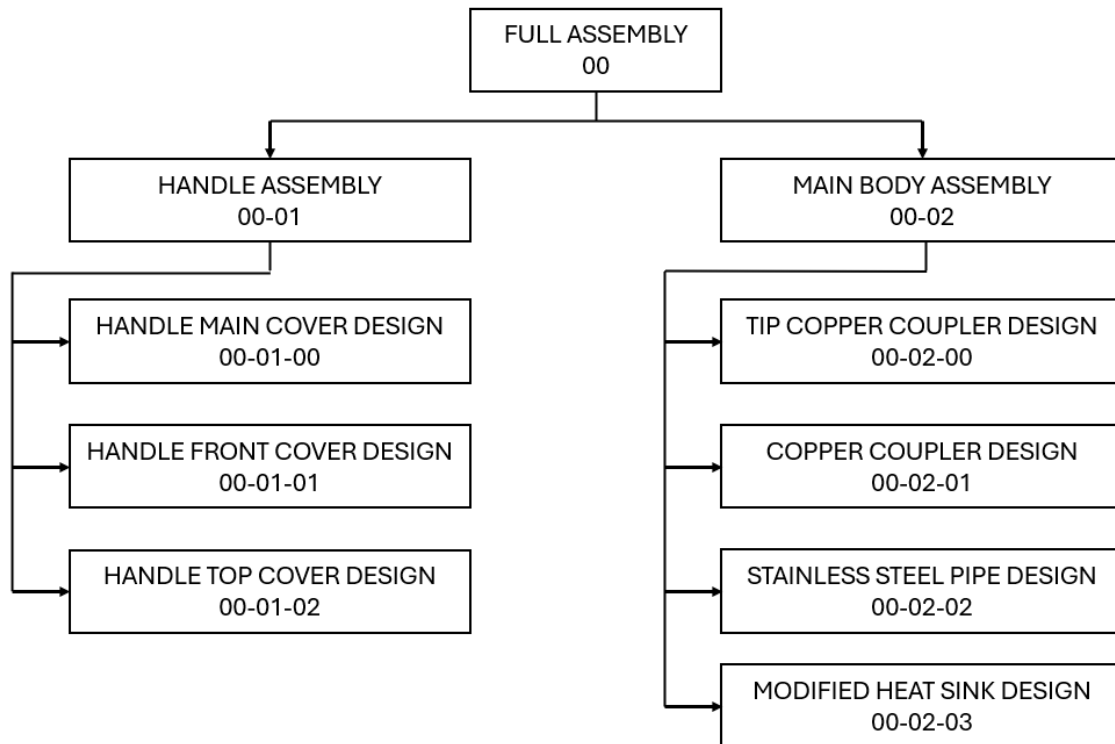


Laparoscope Drawing Pack

	Student	Supervisor
Initials and Surname	PW Breytenbach	Prof K. Schreve
Email Address	26122146@sun.ac.za	kschreve@sun.ac.za
Contact Number	078 780 9388	
Signature		

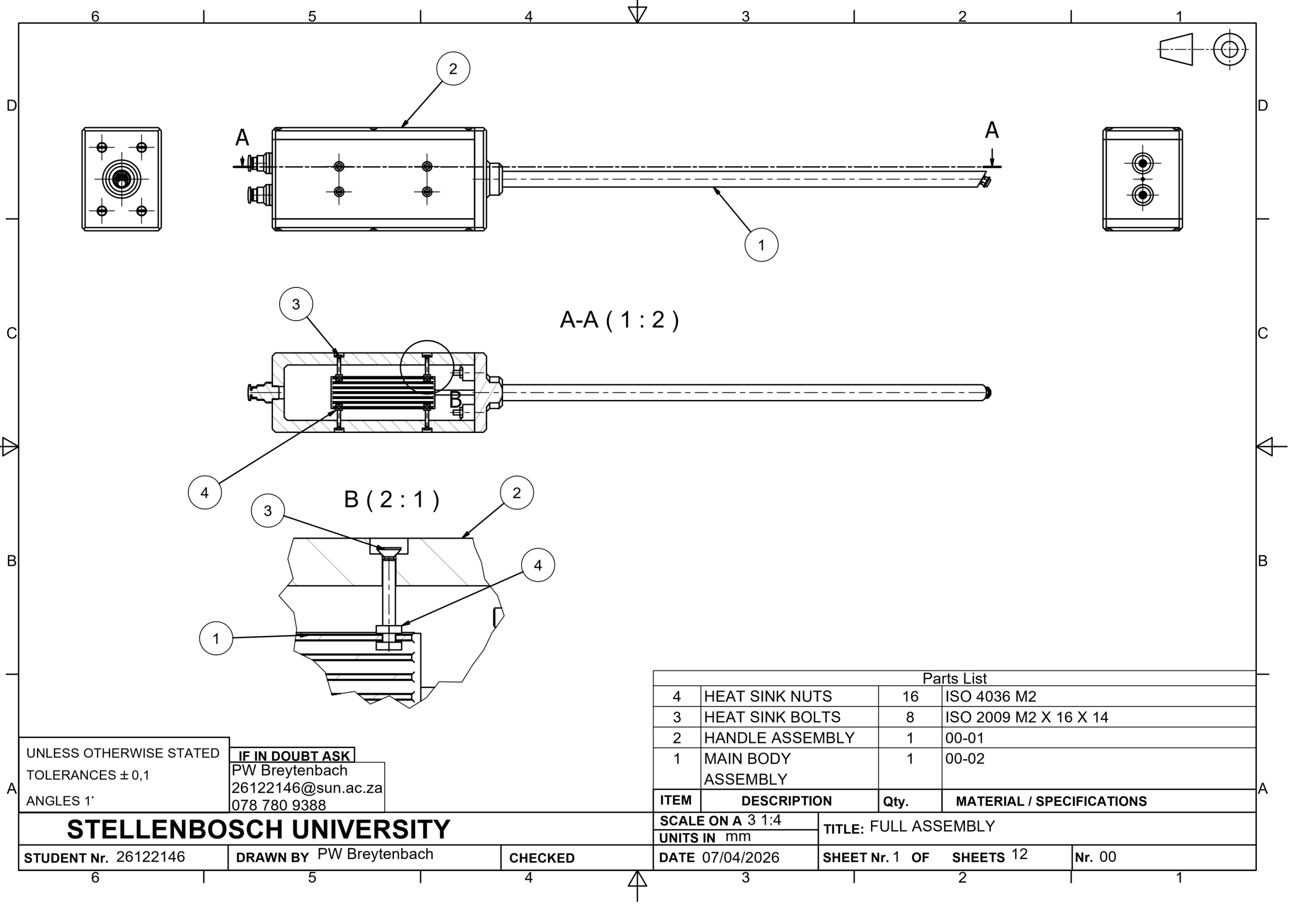


Notes for the assembly:

Parts that interface with each other that are not connected by a bolt and nut connection are secured by epoxy. Thus, all parts that interface with the heat pipes and stainless steel pipe will be secured by epoxy. The entire assembly of the design will be performed by the student, including the application of epoxy.

Order of assembling the handle of the laparoscope:

1. Assemble the entire main body assembly, except for the heat sinks.
2. Insert and secure the heat pipe and stainless steel pipe into the “Handle Front Cover”.
3. Connect the heat pipe to the heat sinks.
4. Connect the “Handle Front Cover” to the “Handle Main Cover”.
5. Connect the heat sinks to the “Handle Main Cover”.
6. Connect the “Handle Top Covers” to the “Handle Main Cover”.

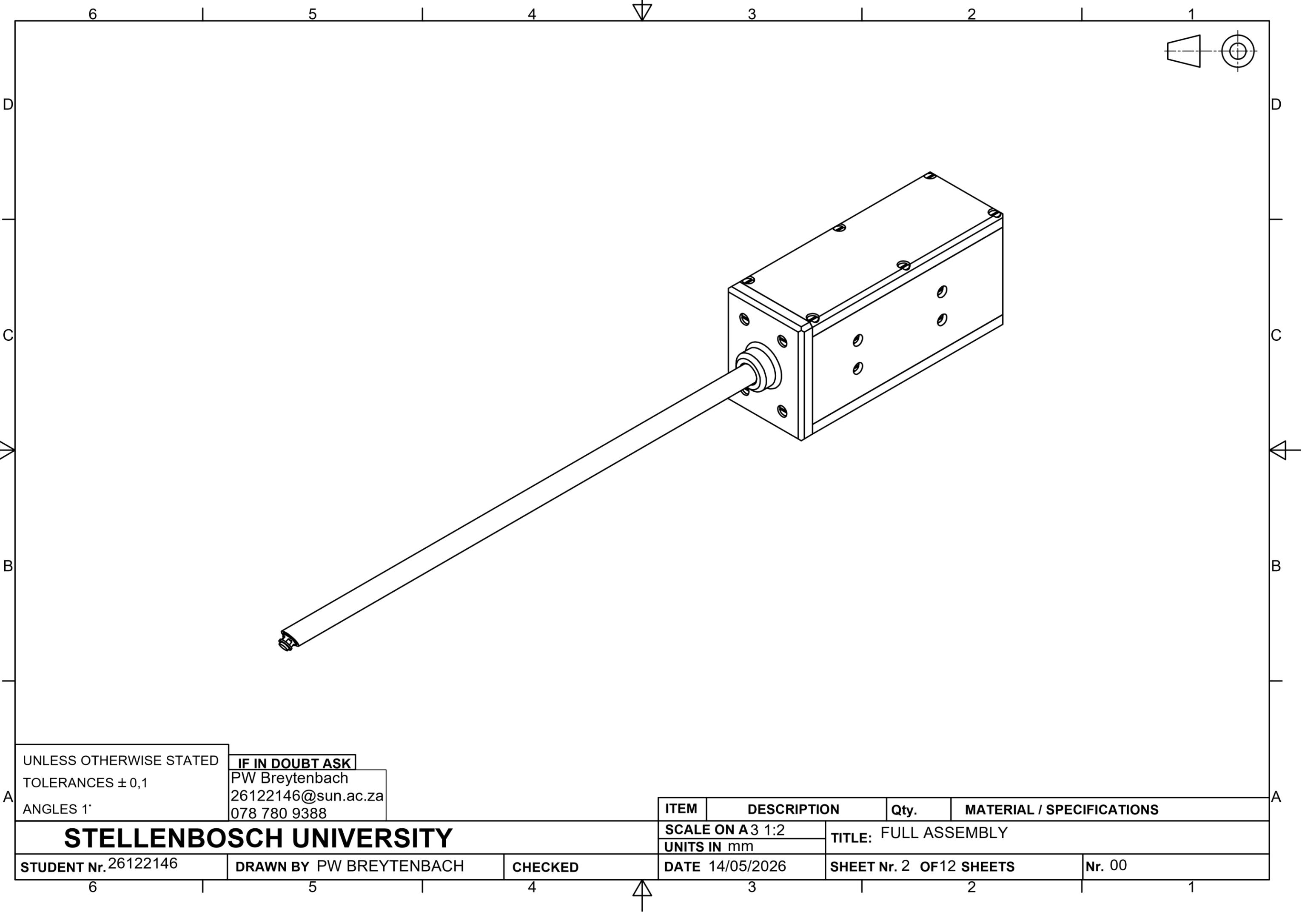


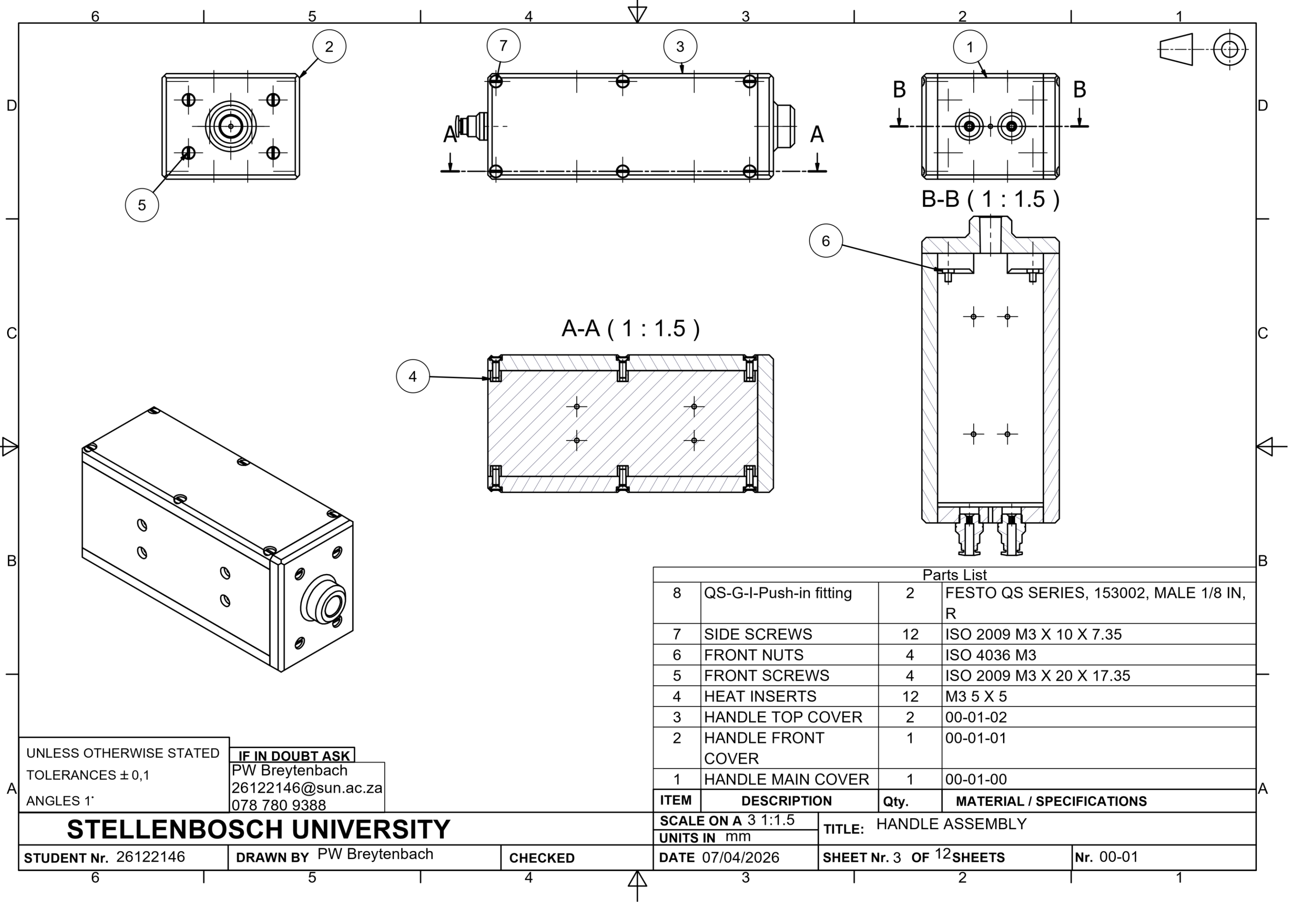
UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

IF IN DOUBT ASK
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Parts List			
4	HEAT SINK NUTS	16	ISO 4036 M2
3	HEAT SINK BOLTS	8	ISO 2009 M2 X 16 X 14
2	HANDLE ASSEMBLY	1	00-01
1	MAIN BODY ASSEMBLY	1	00-02
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS

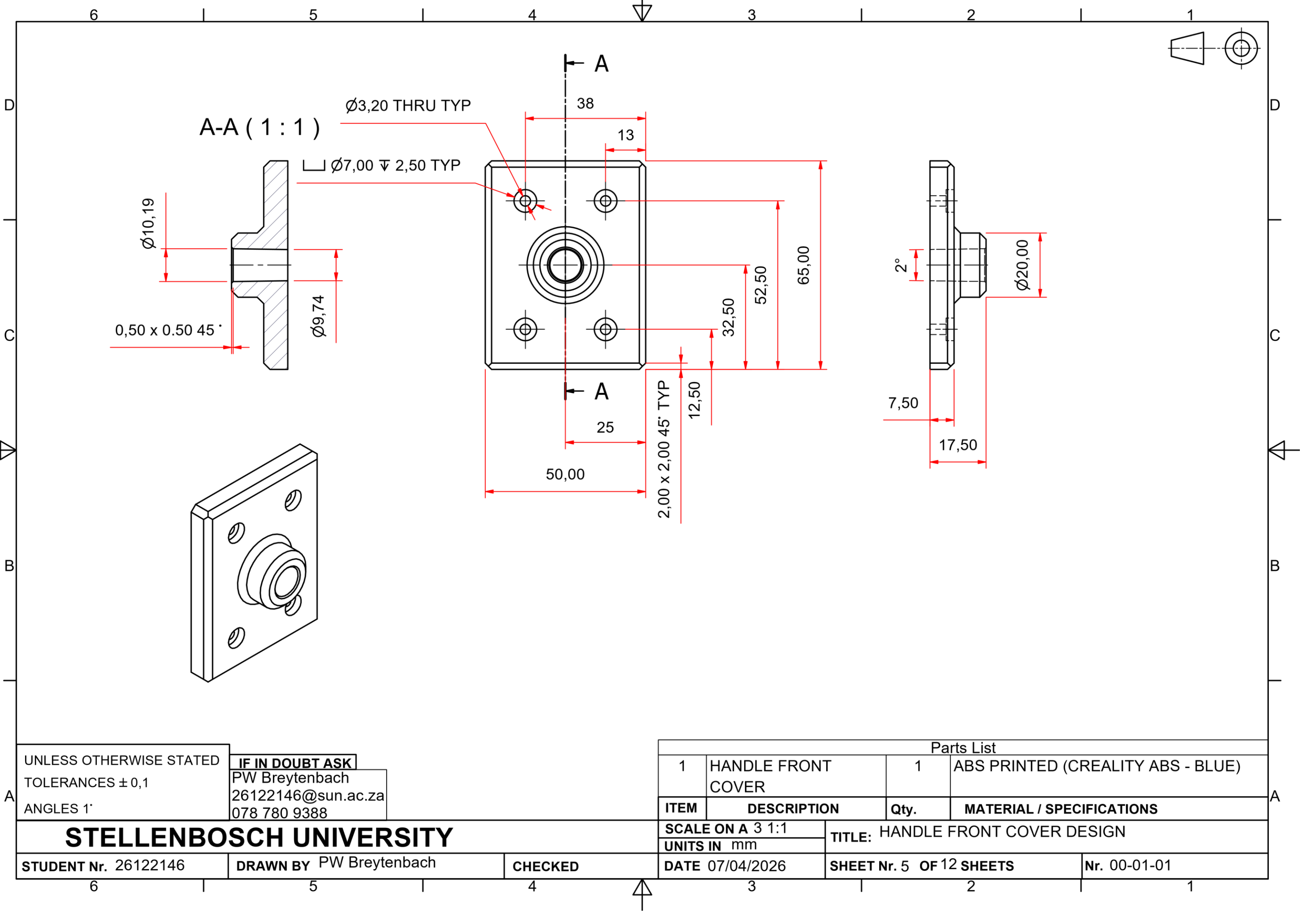
STELLENBOSCH UNIVERSITY			SCALE ON A 3 1:4	TITLE: FULL ASSEMBLY	
			UNITS IN mm		
STUDENT Nr. 26122146	DRAWN BY PW Breytenbach	CHECKED	DATE 07/04/2026	SHEET Nr. 1 OF SHEETS 12	Nr. 00





UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

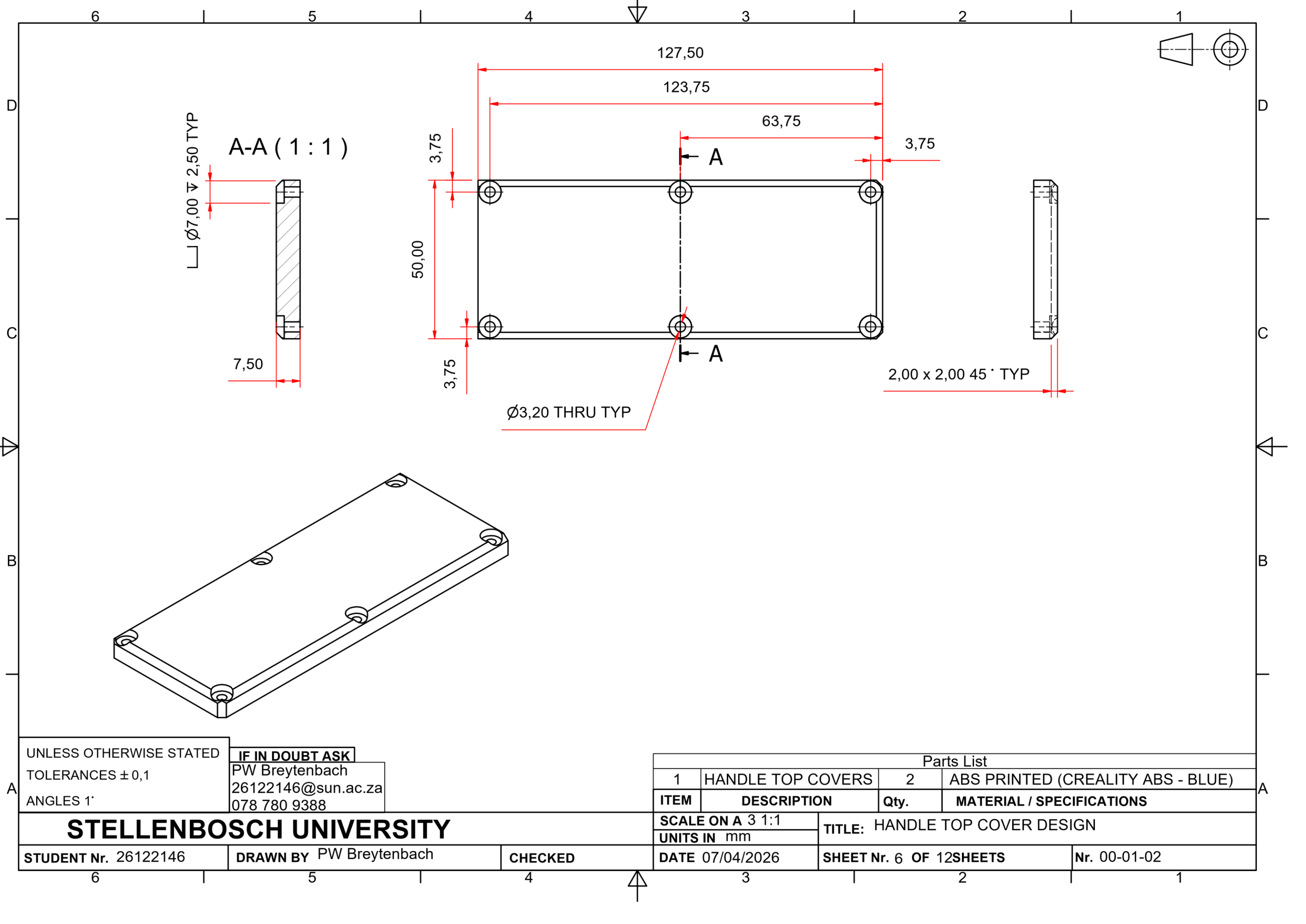
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TOLERANCES $\pm 0,1$
ANGLES 1°

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Parts List			
1	HANDLE FRONT COVER	1	ABS PRINTED (CREALITY ABS - BLUE)
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
SCALE ON A 3 1:1 UNITS IN mm		TITLE: HANDLE FRONT COVER DESIGN	
STUDENT Nr. 26122146		DRAWN BY PW Breytenbach	
DATE 07/04/2026		SHEET Nr. 5 OF 12 SHEETS	
CHECKED		Nr. 00-01-01	

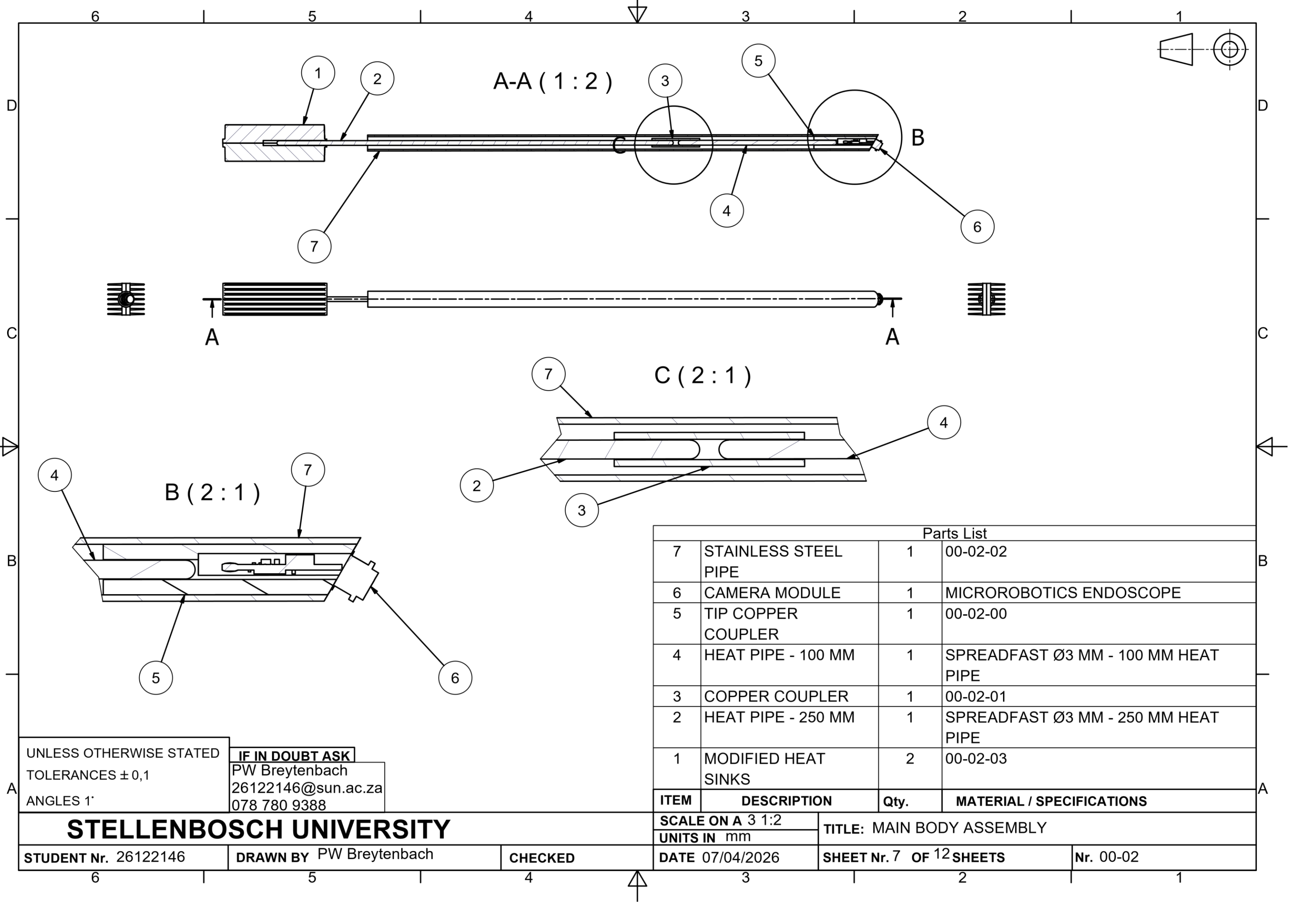


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ANGLES 1°

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STELLENBOSCH UNIVERSITY			
STUDENT Nr. 26122146	DRAWN BY PW Breytenbach	CHECKED	

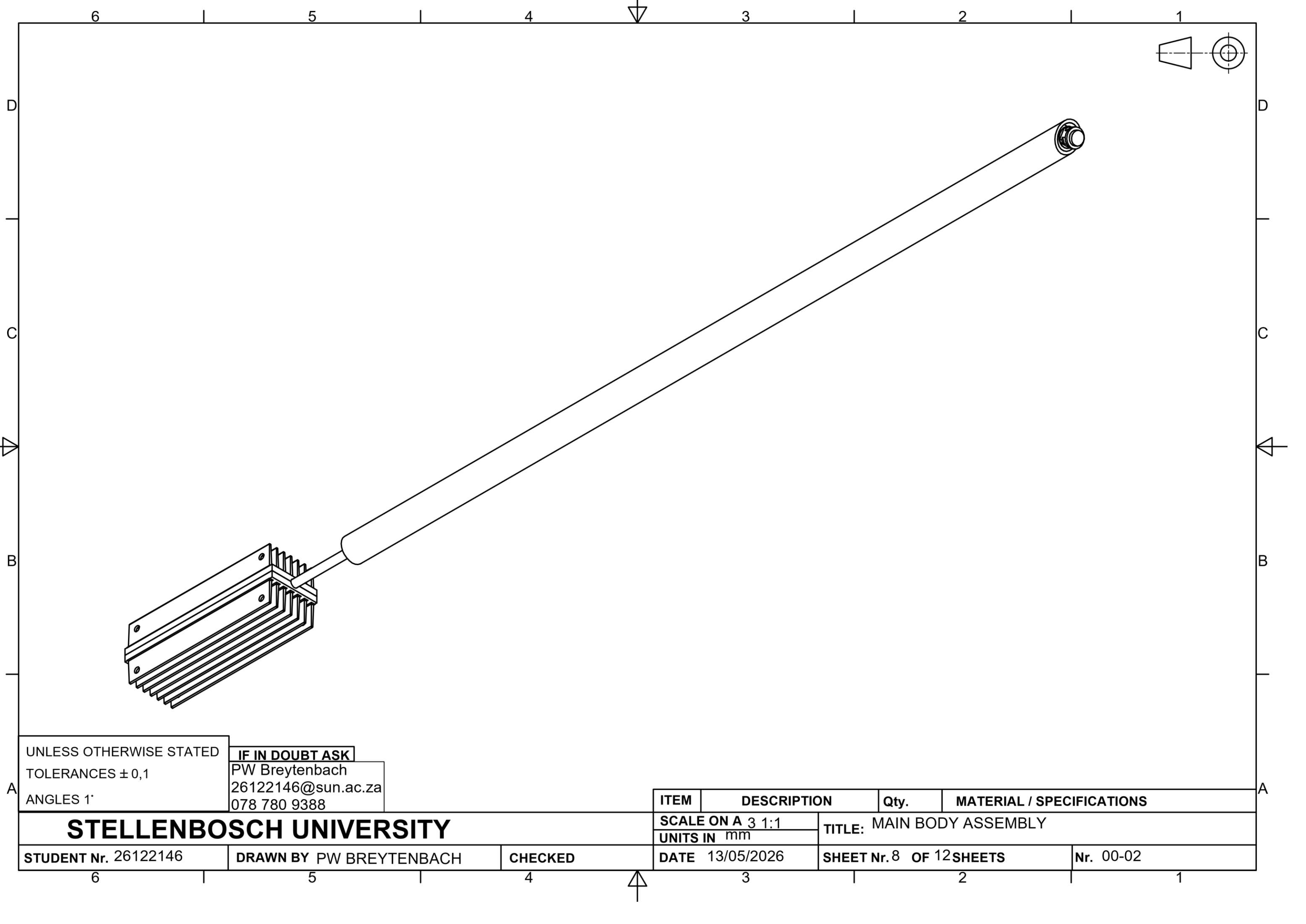
Parts List			
1	HANDLE TOP COVERS	2	ABS PRINTED (CREALITY ABS - BLUE)
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
SCALE ON A 3 1:1 UNITS IN mm		TITLE: HANDLE TOP COVER DESIGN	
DATE 07/04/2026		SHEET Nr. 6 OF 12SHEETS	
		Nr. 00-01-02	



UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

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Parts List			
7	STAINLESS STEEL PIPE	1	00-02-02
6	CAMERA MODULE	1	MICROROBOTICS ENDOSCOPE
5	TIP COPPER COUPLER	1	00-02-00
4	HEAT PIPE - 100 MM	1	SPREADFAST Ø3 MM - 100 MM HEAT PIPE
3	COPPER COUPLER	1	00-02-01
2	HEAT PIPE - 250 MM	1	SPREADFAST Ø3 MM - 250 MM HEAT PIPE
1	MODIFIED HEAT SINKS	2	00-02-03
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS

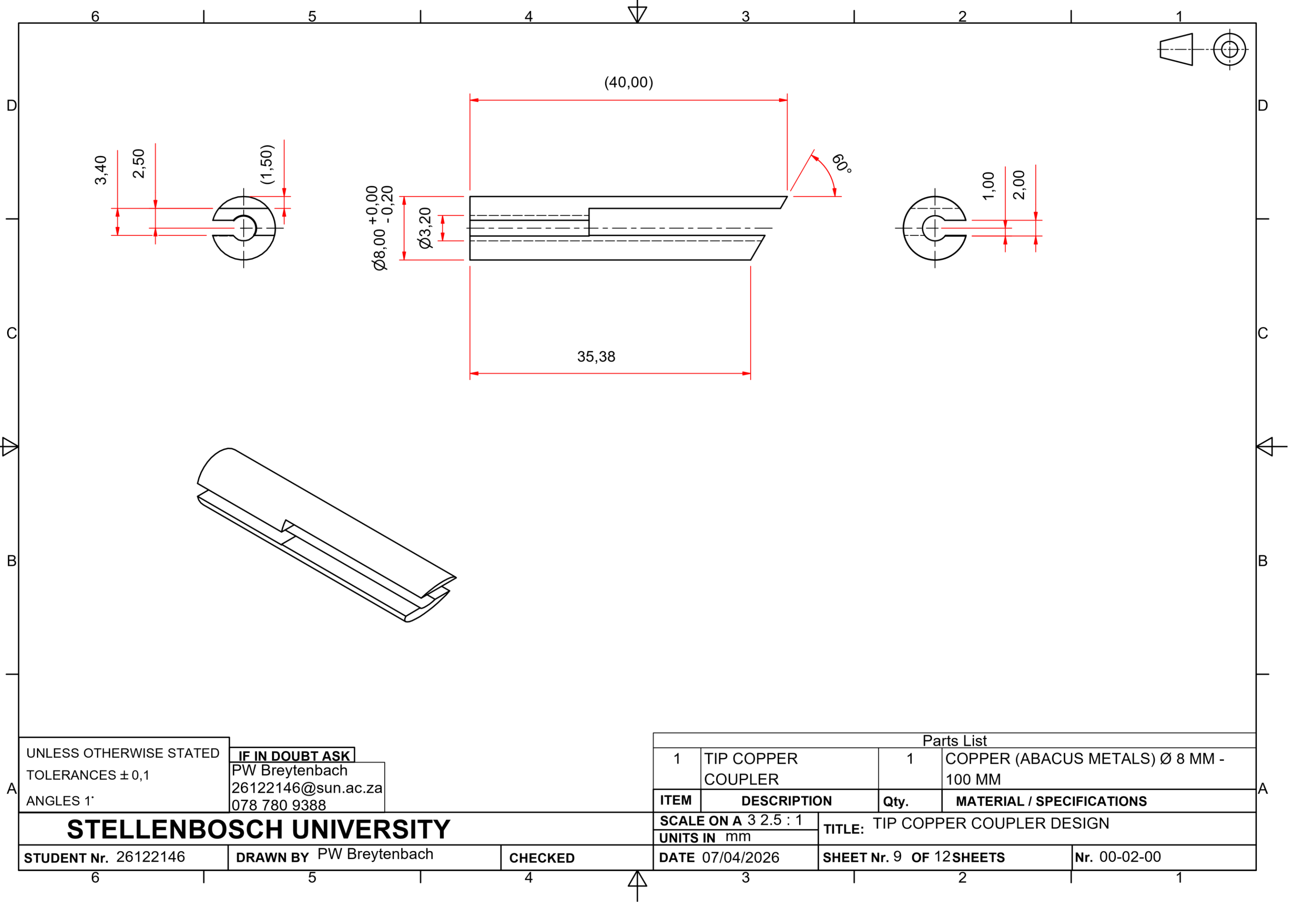


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ANGLES 1°

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STELLENBOSCH UNIVERSITY			
STUDENT Nr. 26122146	DRAWN BY PW BREYTENBACH	CHECKED	

ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS	
SCALE ON A 3 1:1		TITLE: MAIN BODY ASSEMBLY		
UNITS IN mm				
DATE 13/05/2026		SHEET Nr.8 OF 12SHEETS		Nr. 00-02

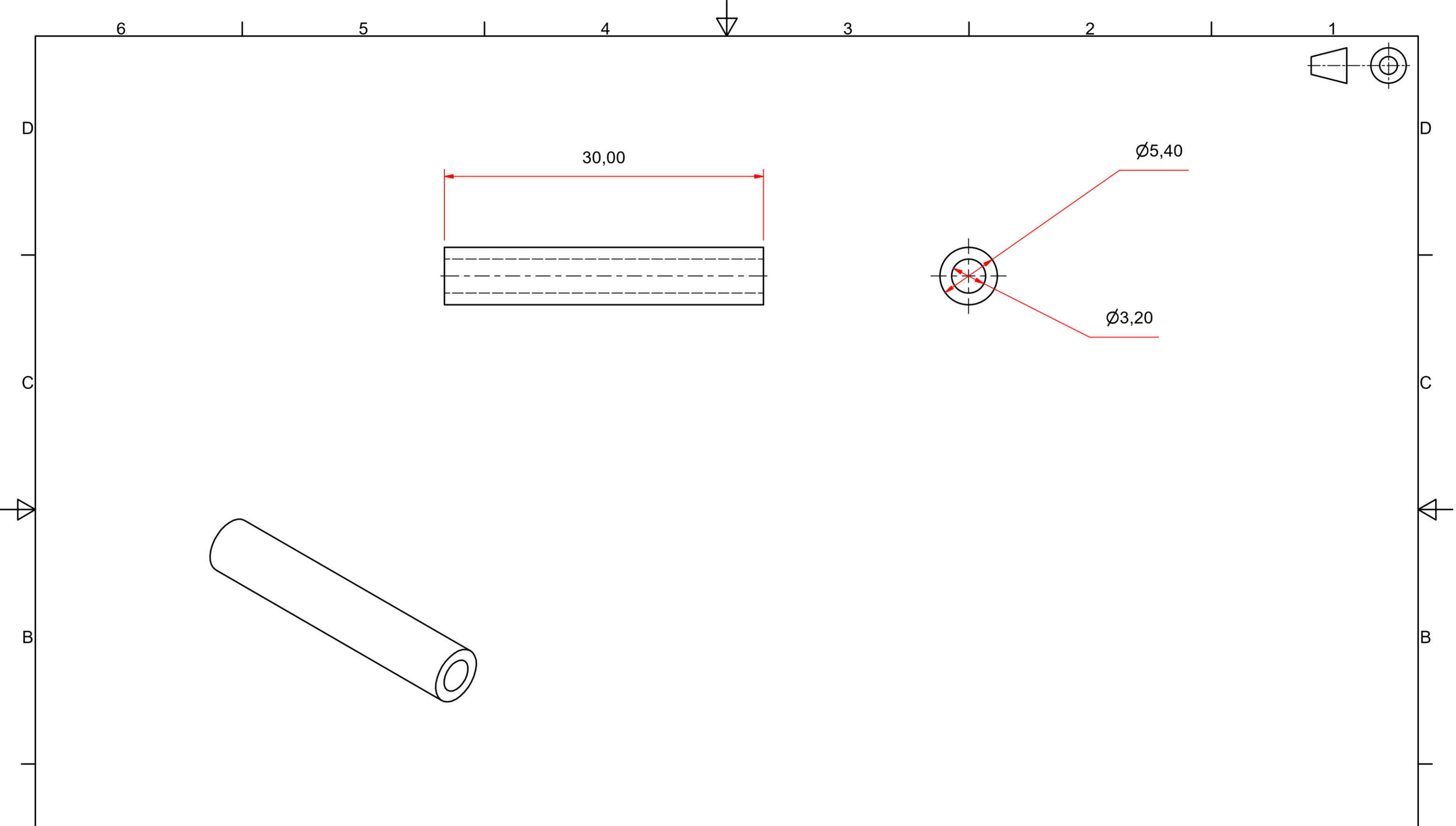


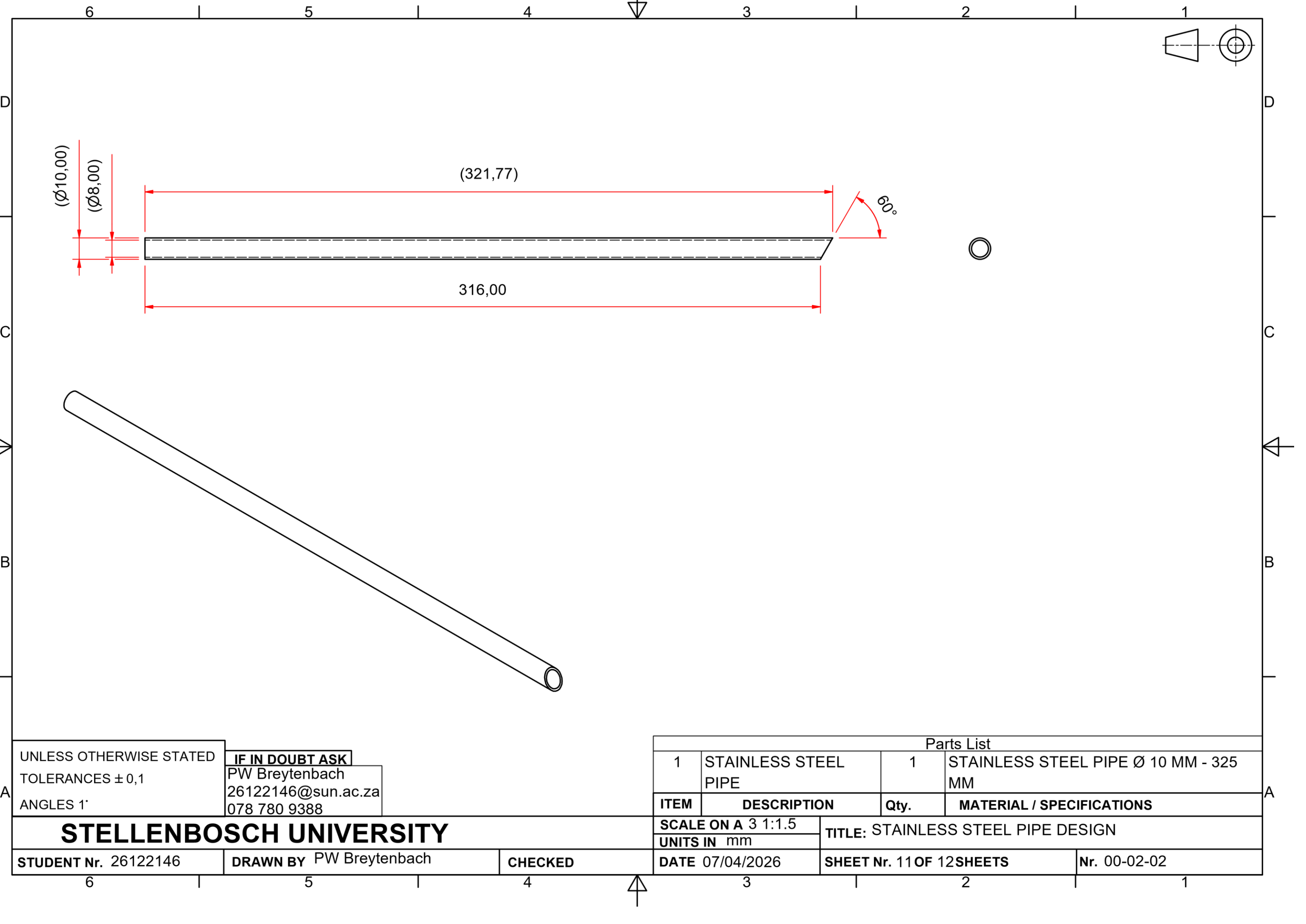
UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

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STELLENBOSCH UNIVERSITY		
STUDENT Nr. 26122146	DRAWN BY PW Breytenbach	CHECKED

Parts List			
1	TIP COPPER COUPLER	1	COPPER (ABACUS METALS) Ø 8 MM - 100 MM
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
SCALE ON A 3 2.5 : 1 UNITS IN mm		TITLE: TIP COPPER COUPLER DESIGN	
DATE 07/04/2026		SHEET Nr. 9 OF 12SHEETS	
		Nr. 00-02-00	



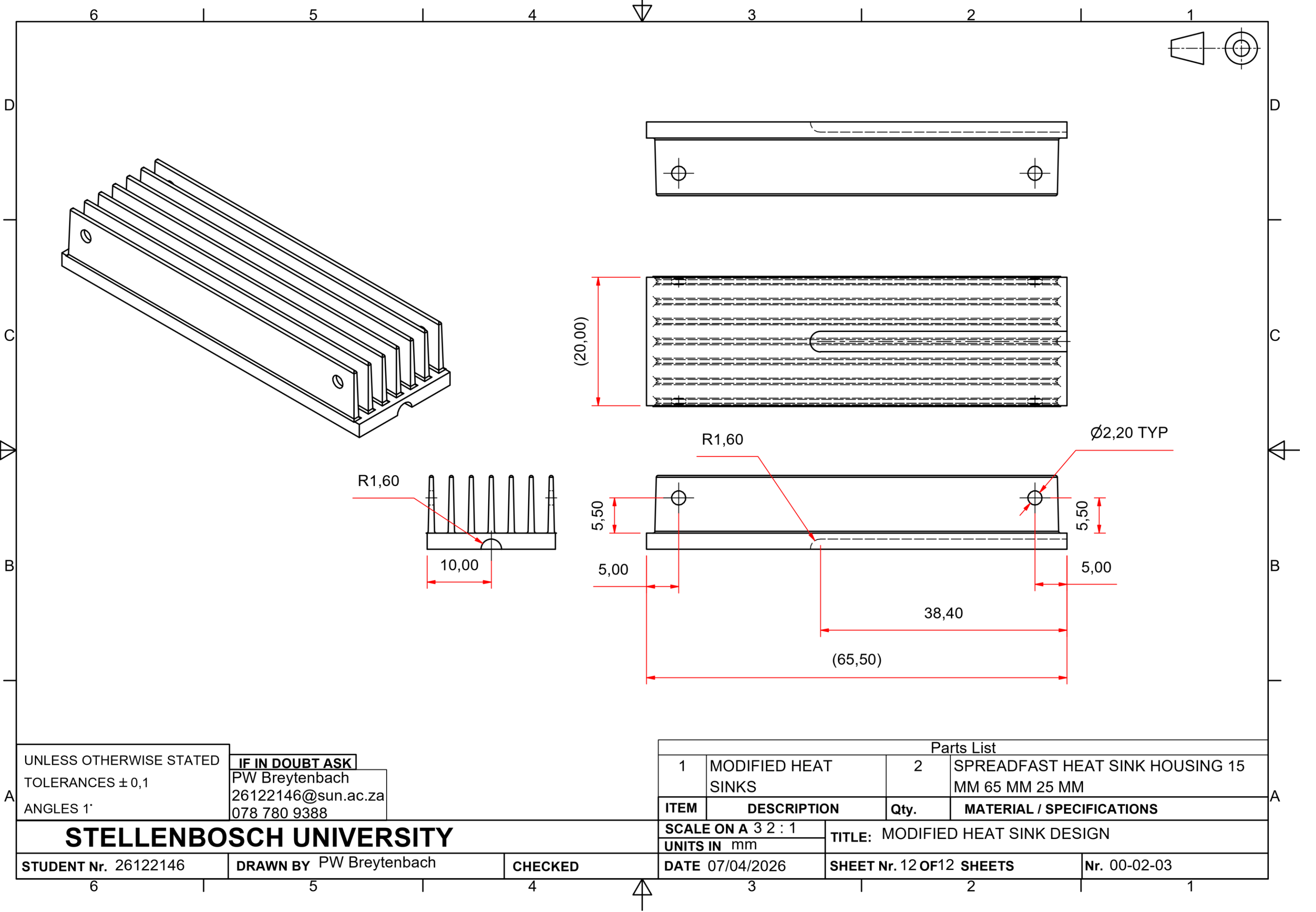


UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

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STELLENBOSCH UNIVERSITY		
STUDENT Nr. 26122146	DRAWN BY PW Breytenbach	CHECKED

Parts List			
1	STAINLESS STEEL PIPE	1	STAINLESS STEEL PIPE Ø 10 MM - 325 MM
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
SCALE ON A 3 1:1.5		TITLE: STAINLESS STEEL PIPE DESIGN	
UNITS IN mm			
DATE 07/04/2026		SHEET Nr. 11 OF 12SHEETS	Nr. 00-02-02



UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

IF IN DOUBT ASK
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Parts List			
1	MODIFIED HEAT SINKS	2	SPREADFAST HEAT SINK HOUSING 15 MM 65 MM 25 MM
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
SCALE ON A 3 2 : 1 UNITS IN mm		TITLE: MODIFIED HEAT SINK DESIGN	
STUDENT Nr. 26122146		DRAWN BY PW Breytenbach	
CHECKED		DATE 07/04/2026	
		SHEET Nr. 12 OF 12 SHEETS	
		Nr. 00-02-03	